Roll No. IC-21421-22 IC-812A

International Institute of Professional Studies
MCA(5Yrs) -VIII Semester
End-Sem Examination April-May, 2025
IC-812A: Soft Computing

Time: 3 Hrs **Max. Marks: 60**

NOTE: All questions are compulsory. Attempt any two parts from each question. All questions carry equal marks. Answers must be brief and to the point. Attempt all the parts of a question on same place. Any assumptions made answering the questions should clearly be stated.

Q.1 (a) Describe the characteristics of soft computing that distinguish it from hard computing. Draw and explain the computational paradigms. 6

(b) What are the main components of soft computing? Explain each component briefly. 6

(c) Discuss the role of fuzzy logic in soft computing and how it differs from Crisp logic systems. Elaborate it with the stopping distance as per speed of bike 6

Q.2 (a) What are the implications of using hedges in fuzzy logic, and how do they enhance the representation of linguistic variables? 6

(b) Describe the concept of 'youthness' and 'healthy' as a fuzzy subset and how it is determined, What is the membership function used? As per the degree of membership

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a = X is Young and X is Healthy
b = X is Young or X is Healthy
c = not X is Young
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Compute the above values. 6

(c) Using intuition and your own definition of the universe of discourse, plot fuzzy membership functions to the following variables:
(i) Liquid level in the tank
(a) Very small
(b) Small
(c) Empty
(d) Full
(e) Very full 6

Q.3 (a) With the help of necessary block diagrams, compare Mamdani and Sugeno Fuzzy Inference Systems. 6

(b) What is the role of the fuzzy inference engine in fuzzy logic systems, and how does it utilize rule aggregation? 6

(c) How do membership functions characterize fuzzy sets, and what are the implications of their properties for fuzzy reasoning? 6

Q.4 (a) What do you understand by artificial neural networks? Explain some of its important applications. 6

(b) What is mean by learning what is supervised and unsupervised learning give example of each. 6

(c) What are different types of neural networks based on architecture? explain 6

Q.5 (a) What is genetic algorithm? explain with flowchart. 6

(b) Describe various operators of genetic algorithm. 6

(c) What is the significance of encoding methods in genetic algorithms, and how do binary and permutation encoding differ in their applications? 6

Roll No. IC-21021-22 IC-812

International Institute of Professional Studies
End Semester Exam April-May, 2025
MCA (5 Yrs) VIII Semester
Theory of Computation

Time: 3 Hrs **Max. Marks: 60**

Note: **Question No. 1** is compulsory. Attempt ***any four*** questions from remaining questions. Attempt subparts of a question together at one place.

1. A) Show that {abc, bca, cab} can be generated by a regular grammar whose terminal set is {a, b, c}. (2×6=12)

   B) Give formal definition of finite automata. Why finite automata are called finite?

   C) Construct a Mealy machine, which can output EVEN, ODD according as the total number of 1's encountered is even or odd. The input symbols are 0 and 1.

   D) Give the minimal DFA that performs as a mod-3 1's counter, i.e. accepts a string each time the number of 1's in the input sequence is a multiple of 3.

   E) Construct a regular expression to represent the set of strings over {a, b} having exactly three 3b's.

   F) Write Church-Turing thesis.

2. A) Write formal definition of grammar. Write a grammar that generates set of all palindrome strings over {a,b}* (2+4)

   B) What do you mean by ambiguous grammar? Prove that the grammar with following production rules is ambiguous: - (2+4)

   $S \rightarrow a \mid abSb \mid aAb$ $A \rightarrow bS \mid aAAb$

3. A) Explain Chomsky Normal Form. Reduce the following grammar to Chomsky Normal Form: (2+4)

   $S \rightarrow aAA \mid bBB$ $A \rightarrow a \mid aA$ $B \rightarrow b \mid bB$

   B) Consider the following grammar G with production rules: (3+3)

   $S \rightarrow AB \mid BC$ $A \rightarrow BA \mid a$ $B \rightarrow CC \mid b$ $C \rightarrow AB \mid a$

   Find using CKY algorithm whether the strings **baaba** and **baaaba** are accepted by the above grammar

4. A) Write the formal definition of Mealy machine and Moore machine. Construct a Moore machine equivalent to the following Mealy machine:- (3+3)

Present State	a=0		a=1	
	State	Output	State	Output
→q1	q1	1	q2	0
q2	q4	1	q4	1
q3	q2	1	q3	1
q4	q3	0	q1	1

P.T.O.

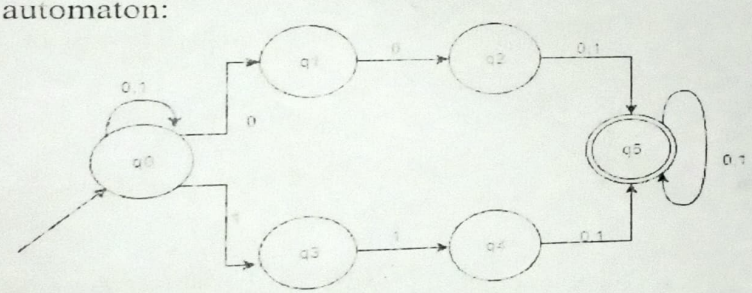
B) Construct a minimum state automaton equivalent to the finite automaton described by the following transition table: (6)

state	Input	
	a	b
→q0	q1	q2
q1	q2	q3
q2	q2	q4
q3	q3	q3
q4	q4	q4
q5	q5	q4

5. A) Construct finite automata equivalent to following regular expressions: (3+3)

i) $(ab + a)^* (aa + b)$ ii) $a (a + b)^* ab$

B) Construct a regular expression equivalent to following finite automaton: (6)



6. A) Write formal definition of Pushdown Automata (PDA). Construct a Pushdown Automata (PDA) for following language: (3+3)

$L = \{ \{a^n b^n \mid n>=1\}$

B) Construct a Pushdown Automata (PDA) for following grammar: (3+3)

$S \rightarrow aAB \mid bBA$ $A \rightarrow bS \mid a \mid aS$ $B \rightarrow b$

Check whether the string **abaabb** is accepted by the PDA. Show stack's contents at each instance.

7. A) Give formal definition of Turing machine. Construct a Turing Machine for the following language :- (2+4)

$L = \{a^n b^n c^n \mid n > 0\}$

Draw transition diagram only.

B) Explain Turing machine with Stay option. How can a Standard Turing Machine simulate the behavior of a Turing machine with Stay option? Explain. (3+3)

Q:1 A)What is the difference between OLTP and OLAP in terms of transactions per day

How many rows will each partition have if 800 million rows are split into 8 partitions?

B)Explain how breaking down large tables into partitions helps performance.

Describe what happens when we roll-up from daily to quarterly data.

C) Calculate the percentage improvement if reporting time goes from 72 hrs to 4 hrs.

If a system processes 500 transactions/sec, how many will it handle in a day?

D) Analyze the network time required to transfer 120 GB at 200 MB/min.

Compare the time taken for 100 OLTP vs OLAP queries.

Q:2 a)What is a star schema?

List the types of OLAP operations.

B) Why is sparsity important in a data cube?

What is the purpose of using a snowflake schema over a star schema?

C) A cube has 3 dimensions: Product (10), Region (5), Time (12). How many cells does it contain?

You slice a cube by Region = "North". How many cells remain?

D) Given 4 dimension tables in a star schema, how many joins are required?

Break down the cost of building 5 departmental data marts, each costing $8000.

Q:3 A)What is support in data mining?

Define classification accuracy.

B)Explain how missing data affects model performance.

Describe the difference between clustering and classification.

C) In a dataset of 500 transactions, {Bread, Butter} appears 120 times. What is the support %?

A classifier gets 180 correct out of 200. What's the accuracy?

D) A KDD process takes 50 hrs, with 15 hrs for data mining. What % is spent on mining?

An algorithm processes 1000 records in 1 sec. Estimate time for 1 million records.

Q:4 A)What is entropy in a decision tree?

What does TF-IDF measure?

B) Explain how term frequency is calculated.

Describe how PageRank uses links to measure importance.

C) Calculate entropy if a dataset has 4 "Yes" and 6 "No" outcomes.

A web page has 800 words; "data" appears 40 times. What is the TF?

D) Evaluate information gain after a split using entropies of subsets.

Find TF-IDF if a word appears 5 times in a document of 100 words, and in 10 of 1000 documents.

Q.5 A)What is temporal mining?

Define spatial density.

B) Why use moving averages in time-series analysis?

What insights can you get from tracking sales patterns over months?

C) Calculate support if a pattern appears in 4 out of 12 months.

What is the Euclidean distance between point A(1,2) and B(4,6)?

D) Given 500 points over 10 km², find density.

Sales data = [100,120,140,160,180]. What is the 3-month moving average at month 4?

International Instituted of Professional Studies

End Semester Exam April-May 2025

MCA(5yrs) VIII Semester

Mobile and Wireless Computing

RG-21-221-22

Time: 3 hrs **Max. Marks :60**

Note: Q1 is compulsory. Attempt any three questions from Q2 to Q6.

Q1	Answer any 12 questions from a) to o) .	12*2=24
a)	Explain inter-symbol interference.	2
b)	What are Cellular systems?	2
c)	List four numbers/ Identities used in GSM.	2
d)	What is COA(care of address)?	2
e)	Explain packet reservation multiple access.	2
f)	What is wireless senor network?	2
g)	Explain quadrature phase shift keying.	2
h)	Write short note on antennas.	2
i)	Compare infra red and radio transmission	2
j)	Compare infrastructure and ad-hoc network	2
k)	Write short note on Internet of everything.	2
l)	Explain Dropper effect.	2
m)	Explain Frequency reuse and its effects	2
n)	Explain cell spitting and its effects.	2
o)	Explain cell sectoring and its effects.	2
Q2 a)	What are the main benefits of spread spectrum system? How can spreading be achieved? How can DSSS system benefit from multipath propagation?	6
b)	Describe the functions of MS and SIM. How and where is user related data presented/stored in GSM system? How is user data protected from unauthorized access, especially over air interface?	6
Q3 a)	Discuss Mater-Slave architecture in piconet of Bluetooth system. What are the differ states in which Bluetooth device can be?	6
b)	Explain how IOT works? List few of its applications.	6
Q4 a)	Describe System architecture of IEEE 802.11.	4
b)	Explain Basic DFWMAC/DCF using RTS and CTS extension	4
c)	Explain MAC frames of IEEE 802.11? Explain interpretation of MAC addresses in 802.11 MAC frame	4
Q5 a)	What is MobileIP? Why it is required?	4
b)	How registration of mobile node is done to through foreign agent and home agent?	4
c)	How to change foreign agent in Optimized MobileIP?	4
Q6 a)	What are special challenges in routing in ad hoc network?	3
b)	Compare DSDV and DSR routing algorithms.	3
c)	Explain and compare indirect TCP and snooping TCP.	6